This article deals with image processing on PandaBoard using Kinect and OpenCV. It covers installing OpenCV with Kinect support on PandaBoard, along with a few examples.

I just finished conducting a tutorial on this topic at LinuxCon North America 2012. Let me first explain the motivation behind choosing this topic. Recently, there have been a lot of cool mind-blowing hacks using Kinect, such as the minority report replica, gesture-detection system, etc. But all of these hacks or systems developed are not at all portable! It would be very hard to even commercialise them. So, why not start developing your idea on a mobile platform such as PandaBoard?

Why are we doing this?

This might be your first question. Here are a few of my answers:

- Using the combined potential of the PandaBoard, OpenCV and Kinect, we can possibly solve a lot of computer vision problems.
- Since PandaBoard is very small, we could possibly end up developing a product that can be carried around by users wherever they go.
- It sounds cool!

A quick recap of PandaBoard and OpenCV

I have covered these topics in previous articles, so I'll request you to refer to earlier issues for details. All I'll say here is that PandaBoard is an open source embedded development board based on the Texas Instruments OMAP 4470 platform, which supports various Linux distributions. Incidentally, its price has now dropped to $153, while the PandaBoard ES version is available for $162.

OpenCV is one of the world's most popular computer vision libraries, free for commercial and research use. The latest version (2.4.2) also has support for Kinect.

Microsoft Kinect

Kinect is a motion-sensing input device made by Microsoft. If you buy Kinect for Windows, you also get a commercial licence with it. The price is $250; for students, it's $150. Read more about it at Wikipedia if you need to.

Installing OpenNI and Kinect drivers (Ubuntu on PandaBoard)

Note: These instructions will only work for Ubuntu running on PandaBoard, and not on a desktop machine or a laptop.

OpenNI (Open Natural Interaction) is a multi-language, cross-platform framework that defines APIs for writing applications utilising Natural Interaction.

The first part, OpenNI installation, begins with installing the dependencies, with the following code:

```
sudo apt-get update
sudo apt-get install gcc-multilib libusb-1.0.0-dev git-core build-essential
sudo apt-get install doxygen graphviz default-jdk freeglut3-dev libopencv-dev
```

Next, create a folder to hold the download and the installation, and download the latest unstable version of the OpenNI software from GitHub:

```
mkdir kinect
```